清华大学能源与动力工程系 2020 春《燃烧理论》
期末考试试卷(A 卷)
Final Term Examination of Combustion (B)

考试时间(Date)： 2020 年 6 月 8 日 班级(Class)：_____

学号(Student No.)：_____ 姓名(Name)：_____

提示 (Instructions):

1 请将你的学生证放在桌子的右上角(Please put your student ID on the desk)

2 考试为**开卷考试**，允许带教科书，作业，课件，笔记本和另外的参考书各一样，但不能互借，答题必须独立完成。(The examination is **OPEN BOOKS**. Calculators, textbook, homework and notebook are allowed, but they cannot be lent to others. You should answer the questions independently and help is not allowed)

3 试卷共有**5**题，你要在**120**分钟内完成这次考试，先检查一下你的卷子，题目是否对，共有**6**页。答案必须写在答题卷上，试题卷上回答无效。答题必须明确，清晰，计算题应给出单位。(There are **5** questions on this exam with **6** pages. You have up to **120** minutes to work on this exam. Please be sure that you have a complete exam package. Please work exam problems on the pages provided. Show your work neatly and write your answers. Give units for all answers requiring them.)

4 如果试卷不对，请立即与监考教师说明。(If a page is missing, take the exam to a proctor immediately.)

5 强烈推荐你先读一下所有的问题，并对回答问题的时间进行安排。答卷完成后请再仔细检查一下。(Budget your time for each question and be sure to first answer those questions that you know how to work. Check your work after completing the exam.)

6 请你现在就在答卷纸和试卷纸上写上你的**名字与学号**，并在你的答卷纸首页上标明你的试卷的类型(A或B卷)(**PRINT your name NOW** in the spaces at the top of your pages. Mark the exam type A or B)

7 答卷可以用中文或英文，如果是双语的，以中文为准。试卷有英译的，也以中文为准。(The answers can be in English or Chinese. If two languages are used in one answer, I will take Chinese as the selection. Some of the problems have been translated in English. If there are any difference between Chinese and English, Chinese is for your choice.)

一、简答分析题(Answer the following questions briefly) [6 题，共 43 分]

1. 请问火焰厚度的物理意义是什么，火焰贝克利数（Pe_f）是如何定义的？有人认为高压下火焰厚度减小，火焰速度也减小，所以火焰 Pe_f 会急剧减小，你如何判断并简要分析。（7 分）

   What is the physical meaning of flame thickness? What is the definition of flame Peclet number Pe_f ? Someone claims that under high pressure, both the flame thickness and flame speed decrease, so too will the flame Peclet number decrease sharply. What is your opinion? Please explain briefly. (7 points)

2. 对于碳颗粒的燃烧，什么是化学动力学控制和扩散控制？将单模模型改成双膜模型，已知碳表面二氧化碳浓度 $Y_{CO2,S}$、无穷远二氧化碳浓度 $Y_{CO2,\infty}$和火焰面二氧化碳浓度 $Y_{CO2,f}$，碳燃烧速率$\dot{m}_c$，请尝试用已知参数写出扩散阻力和化学动力学阻力的表达式。（8 分）

   What is kinetic controlled regime and diffusion controlled regime, respectively, in the combustion of carbon particle? Switching the one-film model to the two-film model, if we know the CO_2 concentration at the carbon surface $Y_{CO2,S}$, at infinitely far area $Y_{CO2,\infty}$, and at flame surface $Y_{CO2,f}$, and the burning rate of carbon $\dot{m}_c$, please formulate the diffusion resistance and chemical kinetic resistance using the known parameters. (8 points)

3. 为了大幅度降低 NOx 排放并减少对环境的影响，一个民用航机的发动机采用了超贫燃预混燃烧模式，尝试判断与常规方式相比，其发动机中燃料(航空煤油)点火会变得更容易还是更困难？火焰稳定性又如何变化？请解释理由。（7 分）

   To mitigate the NO_x emission and the adverse effect on the environment, a civil aircraft engine applies the ultra-lean premixed combustion mode. Give an estimation on whether the ignition of the fuel (kerosene) becomes easier or harder as compared to the normal combustion mode. How about the flame stability? Please explain in detail. (7 points)

4. 一个正己烷(C6H14)液滴在 100℃，1 atm 环境中燃烧，如果环境介质从空气变成纯氧，其传递数如何变化？而液滴燃烧速率大体上能变化多少倍？（6 分）

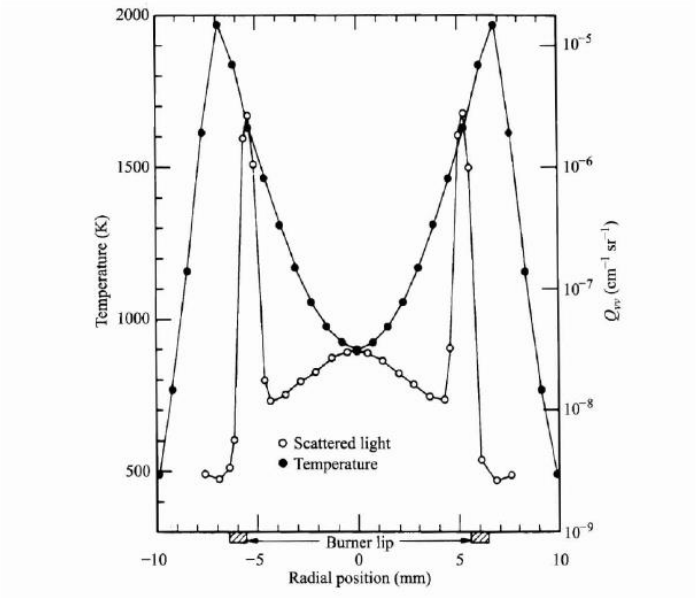
（正己烷燃烧焓 49696 kJ/kg，蒸发焓 335 kJ/kg，气体比热取 1 kJ/kgK，忽略液滴热惯性，沸点 69℃）

A n-hexane (C_6H_{14}) droplet burns under 100 ºC and 1 atm. If the ambient gas changes from air to pure oxygen, how will its transfer number change? And how much times will the burning rate change approximately? (6 points)

(n-hexane: enthalpy of combustion 49696 kJ/kg, enthalpy of evaporation 335 kJ/kg. Specific heat of gas can be taken as 1 kJ/(kg·K). Neglect the thermal inertia of droplet. Boling temperature of n-hexane 69 ºC)

5. 非预混火焰会产生大量的碳烟，请简述碳烟大致生成的途径是什么？如下图，碳烟的峰值通常分布在扩散火焰面的内部未燃区，而不是外部的燃尽区，请分析其原因是什么？（7 分）

Non-premixed flame will produce large amount of soot. Please briefly describe the pathway of soot formation. As in the following figure, the soot concentration usually peaks inside of the diffusion flame surface (the unburned area), not outside of the flame surface (the burned area). What is the cause? (7 points)



6. 火焰弗劳德数 Fr_f 的物理意义是什么？其定义式如何表示，并说明定义式中每一项的物理意义。请问太空近零微重力实验中 Fr_f 数值是多少？（8 分）

What is the physical meaning of flame Froude number? Write down the definition, and explain the physical meaning of each term in the expression. What is value of the Froude number in the microgravity experiments in the space? (8 points)

（12 分）二、室温下通过圆形喷口产生非预混火焰，喷口直径 10mm，出口处气体流速 0.1m/s。若燃料是乙烯和丙烷的混合气，两者体积比为 1:1。试计算火焰高度。若燃料与空气部分预混，燃料空气体积比为 4:1，保持出口处气体流速不变，则火焰高度会变为多少？

(12 points) A non-premixed flame is produced by a circular nozzle at room temperature. The diameter of the nozzle is 10 mm, and the speed of the gas at the nozzle exit is 0.1m/s. If the fuel is a mixture of ethylene and propane with volume ratio 1:1, please calculate the flame length. If the fuel is premixed with air (with the volume ratio of fuel and air being 4:1), keeping the gas speed at the nozzle exit as 0.1 m/s, what will be the new flame length?

（14 分）三、使用定容球形火焰法测量火焰传播速度，在直径 1m 的封闭球壳中充满预混气体，在正中使用电火花点火，火焰从内向外传播，可以认为未燃气体被绝热压缩。球壳上有一压力传感器，记录未燃气体的压力变化。使用高速相机拍摄火焰面图像并计算火焰面移动的速度，拍摄帧率为 1000 帧每秒。实验中，当火焰面直径为圆球直径一半时，记录到两帧之间火焰面直径变化为 1.6mm，两帧拍摄时对应的压力分别为 304.06kPa 和 304.32kPa。试推导火焰传播速度的表达式并计算此时的火焰传播速度（比热比 γ=1.4）。

(14 points) Flame speed is measured by the constant volume spherical flame method. A spherical shell with diameter 1m is filled with premixed gas which is ignited at the center with a spark. The flame propagates outwards, and the unburned gas is adiabatically compressed. A pressure sensor is set on the shell to measure the pressure of the unburned gas. A high-speed camera is used to record the image of flame front and calculate the speed of its movement at a frame rate of 1000 frame per second. In the experiment, when the diameter of the flame front reaches half of the diameter of the shell, it is recorded that the flame front moved by 0.8 mm between two consecutive frames, and the corresponding pressures of the two frames are 304.06 and 304.32 kPa. Please derive the expression of the flame speed and calculate the flame speed at this time (specific heat ratio γ=1.4)..

（14 分）四、已知一直径为 100 微米的正庚烷（C_7H_{16}）液滴在常压（1atm）空气环境下、500 K 的一个燃烧室中完全燃烧，生成水和二氧化碳。液汽分界面处的燃烧蒸气分压为：$P_{F,s}$ (atm) = A exp (−B/T_s(K))，其中 A = 23340，B = 2879。此时，液滴表面温度并不等于此压力下正庚烷的沸点温度 371.5 K。试求解燃烧时液滴的表面温度、燃烧速率和火焰温度。（要求：从沸点温度启动迭代，液滴表面温度的前后两次迭代温度差小于 5 K 即可）

已知条件：忽略对液滴的加热，正庚烷摩尔质量 100.20 kg/kmol，空气摩尔质量 28.85kg/kmol，产物的平均摩尔质量 28.56 kg/kmol。液滴密度 ρ_l=615 kg/m^3，汽化潜热 h_{fg}=316 kJ/kg，燃料热值为 Δh_c = 44926 kJ/kg。燃料蒸气比热容为 $c_{p,F}$=5.1 kJ/(kg·K)，空气比热容 $c_{p,air}$=2.2 kJ/(kg·K)，假定比热容不随温度变化。气体导热系数请根据$k \propto T^{0.5}$外推得到，其中 1000 K 下氧化剂导热系数为 0.067 W/(m·K)，燃料导热系数为 0.097 W/(m·K)。（提示：对于气体比热容 c_p 的选取，在计算燃烧速率时使用 $c_{p,F}$，而计算火焰温度时使用 $c_{p,air}$； 利用表面温度、表面燃料质量分数和传递数三者的关系）

(14 points) An n-heptane (C_7H_{16}) droplet with a diameter of 100 μm is completely burned in a combustion chamber at 500K and 1 atm in air atmosphere, forming water and carbon dioxide. The partial pressure of the fuel vapor at the liquid-vapor interface is $P_{F,s}$ (atm) = A exp (−B/T_s(K)), where A=23340 and B=2879. The surface temperature of the droplet is not equal to the boiling point temperature, which is 371.5 K under 1 atm pressure. Try to solve the surface temperature, burning rate and flame temperature of the droplet during combustion. (Requirement: Start the iteration using the boiling point temperature. The temperature difference between the two iterations of the surface temperature of the droplet should be less than 5 K)

Known conditions: Neglect the heating process of the droplets; Molecular weight of n-heptane 100.20 kg/kmol; Molecular weight of air 28.85 kg/kmol; Average molecular weight of the product 28.56 kg/kmol. The droplet density ρ_l=615 kg/m^3, latent heat of vaporization h_{fg}=316 kJ/kg, fuel value Δh_c = 44926 kJ/kg. The specific heat of fuel vapor $c_{p,F}$=5.1 kJ/(kg·K), and the specific heat of air $c_{p,air}$=2.2 kJ/(kg·K). Neglect the temperature dependence of specific heat. The thermal conductivity k of oxide and fuel at 1000 K is 0.067 W/(m·K) and 0.097 W/(m·K), respectively, and the values at other temperatures should be extrapolated by $k \propto T^{0.5}$. (Hint: for the determination of c_p, c_{pF} and $c_{p,air}$ should be used in the calculation of burn-out time and the flame temperature, respectively. Use the relations among surface temperature, surface mass fraction of fuel and transfer number.)

（17 分）五、在体积比为 3:7 的氧气氮气混合气中进行碳燃烧实验。实验中，采用三色法测得一直径 200 微米碳颗粒的表面温度为 1873K。此时实验压力为 1 atm。假设碳表面混合气体的平均摩尔质量为 30 kg/kmol，用氮气中二氧化碳的扩散系数代表在实验中的多组分扩散系数。在 293K 时，氮气中二氧化碳的扩散系数为 1.63×10^{-5} m^2/s。表面碳反应速率常数可以 $k_c = 4.016\times10^{8} Exp\left(\frac{-29790}{T_s(K)}\right)$ (m/s) 估算。试采用单膜模型求解

（一）扩散阻力和化学反应动力学阻力（两次迭代的相对误差小于 5%即可）

（二）碳表面的二氧化碳质量分数

(17 points) The carbon combustion experiment is performed in the atmosphere of O_2-N_2 mixture with a volume ratio of 3:7. A carbon particle with a diameter of 200 μm features a surface temperature of 1873 K as measured by the three-color pyrometry. The experimental pressure is 1 atm. Assume that the average molecular weight of the gaseous mixture at the carbon surface is 30 kg/kmol, and the diffusion coefficient of carbon dioxide in nitrogen is used for the multicomponent diffusion coefficient in the experiment. At 293K, the diffusion coefficient of carbon dioxide in nitrogen is 1.63×10^{-5} m^2/s . The rate constant of the surface carbon reaction can be estimated by $k_c = 4.016\times10^{8} Exp\left(\frac{-29790}{T_s(K)}\right)$ (m/s). Try to use the one-film model to solve

(i) The diffusion resistance and the chemical kinetic resistance (The relative error between two consecutive iterations must be less than 5%).

(ii) The mass fraction of carbon dioxide mass fraction at the carbon surface.